Tancrède AMIEL

LinkedIn | GitHub | **Portfolio:** tancredea.github.io

Research interests: mathematics for ML (probability, statistics, optimisation) — random matrices and tensors — quantum-inspired ML

EDUCATION

CENTRALESUPÉLEC — Plateau de Saclay, FR
Diplôme d'ingénieur (Master's-level degree), General Engineering — French Grande École — *2024 – 2028*
GPA: 3.94/4.33 (Year 1), 4.03/4.33 (Year 2) — Gap year 2026–27 (industry placement in Tokyo)
Coursework: **Probability, Statistics, Machine Learning, Convex Optimisation & Duality, PDEs, Wave Control (applied functional analysis), Information Theory, Quantum Physics & Quantum Computing, Algorithms, HPC**

COLLÈGE STANISLAS — Paris, FR
CPGE, PCSI then PC (advanced stream) — intensive undergraduate mathematics and physics* — *2022 – 2024*

RESEARCH & ACADEMIC PROJECTS

EMBEDDED IMITATION LEARNING – AutoGear — CentraleSupélec
Course: Entrepreneurship Track — Grade: 17/20 (A+) — *Jan. 2025 – Oct. 2026*
Automatic derailleur kit for any bike, built from scratch to a working prototype, designed to cost 10× less than the market

- Training data: **real public rides reconstructed into the exact signals the device acquires live, with riders' gears recovered by Viterbi decoding**
- Model: **imitates the rider, predicting from cadence and motion the gear wanted 3 s ahead to statistically compensate for the removed torque (100 boosted trees and hysteresis control, 15 KB, sized for the microcontroller)**
- Result: **in closed-loop replay on the same riders, within one gear of the rider's choice 87% of the time, more often than a torque-based rule controller (81%)**

VARIATIONAL TENSOR NEURAL NETWORKS – Critical analysis — CentraleSupélec
Course: Information Theory & Quantum Information — individual report and oral presentation — *June 2026*
Critical reading of Jahromi & Orús (*Scientific Reports*, 2024) on quantum-inspired deep learning, separating contribution from claims

- Prior art: **SVD-based weight compression and tensorised layers, traced back to 2015**
- Genuine novelties: **hybrid deep networks with MPO layers, a custom DMRG-like variational training algorithm, and entanglement entropy as an interpretability metric accessible during training**

BAYESIAN OPTIMISATION & HPC – Blast-aware routing — CentraleSupélec – CEA-DAM
Course: Optimisation of Infrasonic Wave Detection for CTBT Verification — Grade: 19/20 (A+) — *Feb. – Apr. 2026*
Industrial project with the CEA: minimum-damage routing of a device that may detonate, when a full 3D danger map costs 5,000 h

- Derived objective: **with a constant detonation rate, expected damage reduces to the route's cumulative danger**
- Bayesian optimisation to save time: **a Gaussian process (Matérn 5/2) corrects a fast Kingery–Bulmash prior against the ARMEN code using only the simulations that change the route (path-coupled acquisition), with a stopping rule on the route's cumulative uncertainty; 19 runs out of 10,000 candidates in 2D, uncertainty 14.5% → 4.0%**
- Turn-aware exact search: **A* on (node, heading) states keeps turn penalties additive and the search optimal; parallel Δ-stepping (MPI) on the TGCC supercomputer**

PDE-CONSTRAINED OPTIMISATION – Acoustic absorber placement — CentraleSupélec
Course: Wave Control (applied functional analysis) — Grade: 15/20 (A) — *Nov. 2025*
Minimising a hall's acoustic energy with limited absorbing material: Helmholtz problem, complex Robin condition, pre-fractal wall

- Functional analysis (well-posedness): **non-coercive weak form, beyond Lax–Milgram, treated as invertible plus compact (Fredholm alternative), with uniqueness via unique continuation**
- Optimal control (affordable gradient): **adjoint state giving the Fréchet derivative on the whole wall for two PDE solves**

COMBINATORICS RESEARCH – French Tournament of Young Mathematicians — Animath – TFJM^2
National research competition for secondary-school teams — 3rd place, regional tournament — *Jan. – May 2022*

- Proving optimality: **against an adversary, each yes/no answer at best halves the plans still possible, a floor then met exactly by the idea of numbering every plan in base 2 and asking for its digits**

PROFESSIONAL EXPERIENCE

AI / ML ENGINEER (INTERN) – AI Solution Design & Deployment — Tokyo, JAPAN
Dassault Systèmes K.K. — AI team — *Sept. 2026 – Feb. 2027*

- Generative AI: **RAG pipelines, knowledge graphs and MCP servers on open-weight models; GPU hosting sizing**

DEPUTY PROJECT MANAGER & SOLUTION ARCHITECT – IAgriM — Paris – Bamako, MALI
Danew Talla Electronics — *Jul. – Aug. 2024*

- Offline voice AI for Malian farmers: **speech-to-text, spoken-language model and local RAG base; approved by two ministers, consortium with CIAR and AgriCap**. On hold since September 2024.

TEACHING

ORAL EXAMINER (COLLEUR) IN MATHEMATICS – PC* AND PCSI — Collège Stanislas, Paris
Weekly oral examinations at the board in preparatory classes — *June 2025 – June 2026*
About 50 students examined on undergraduate-level mathematics, 2–4 h per week

PRIVATE TUTOR IN MATHEMATICS AND PHYSICS — Paris, FR
10+ students, from middle school to preparatory classes — *Sept. 2024 – June 2026*
Long-term follow-up, including a PC* student through to his engineering-school admission

SKILLS

Languages	French (native) — English (C1/C2, bilingual) — Spanish (B2, DELE certified)
Programming	Python (NumPy/SciPy, Pandas), LaTeX, MPI/OpenMP, SLURM, Linux, Git, Docker; basics: C++, JavaScript
ML	scikit-learn, TensorFlow, Keras, OpenCV, librosa